

DAB_3k3W – 3.3 kW BIDIRECTIONAL DUAL ACTIVE BRIDGE – LAB PROTOTYPE POWER BOARD

Sheets connect through global labels (V1+, PGND1, V2+, PGND2, A, B, C, D, G1..G8, S1..S8, PWM1..8, +5V, +3V3, GND ...).

Connectors



File: Connectors.kicad_sch

Bridge1



File: Bridge1.kicad_sch

GateDrive



File: GateDrive.kicad_sch

Bridge2



File: Bridge2.kicad_sch

Sensing



File: Sensing.kicad_sch

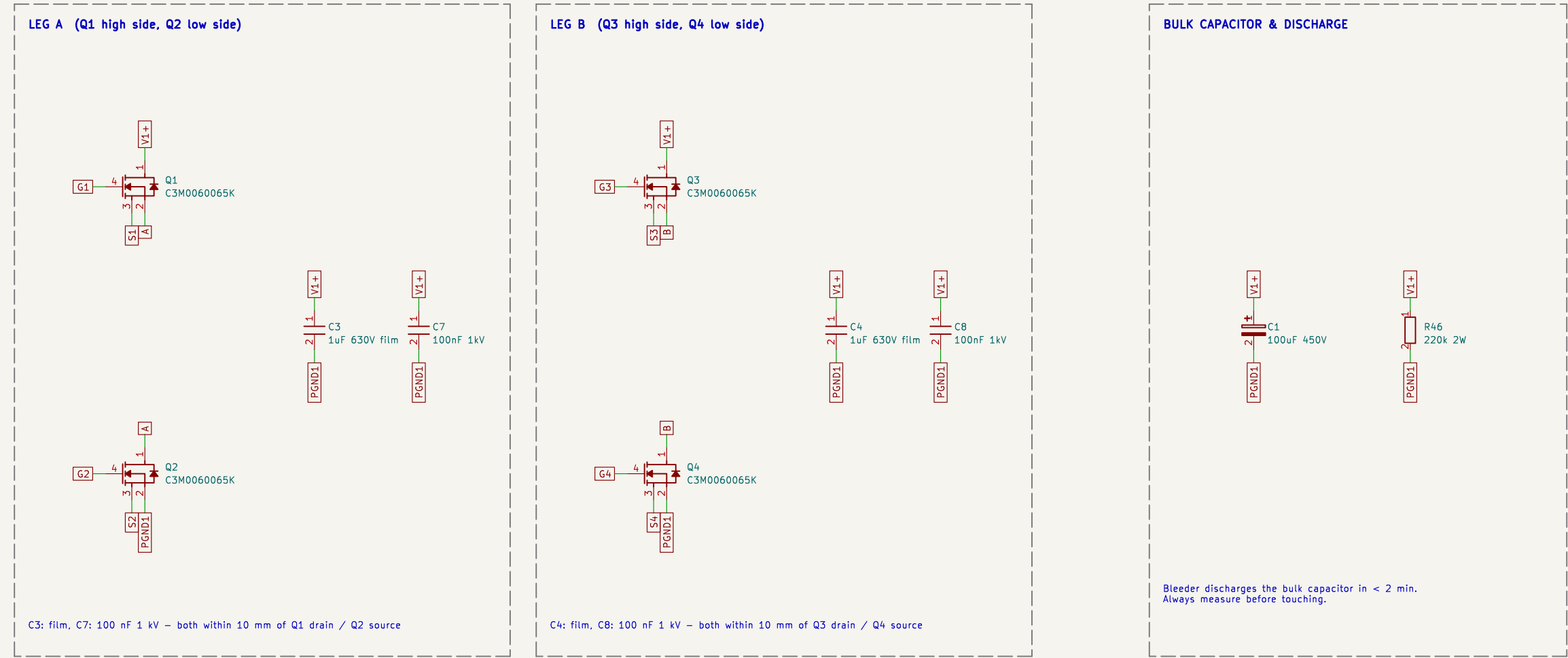
THREE ISOLATED DOMAINS (no copper crosses between them):
PRIMARY 400 V : Bridge 1, C1, DC-bus terminals, ground net PGND1
SECONDARY 350 V : Bridge 2, C2, battery terminals, ground net PGND2
CONTROL 3.3/5 V : J9, logic side of drivers and sensors, ground net GND
Bridged only by: transformer (off-board), UCC21520 drivers, MGJ2 supplies, AMC1311s, LTSR sensors.
PCB: >= 8 mm creepage + 2 mm routed slot under every isolator; >= 3 mm between HV nets.

DESIGN DATA (from the project report)
V1 = 400 V DC bus V2 = 350 V battery (300 V min) P = 3.3 kW
fs = 20 kHz n = N1:N2 = 8:7 Lk = 291 uH (off-board)
SPS: $P = n \cdot V1 \cdot V2 \cdot \phi_i \cdot (\pi_i - |\phi_i|) / (2 \cdot \pi^2 \cdot f_s \cdot L_k)$; $\phi_i > 0$: G2V, $\phi_i < 0$: V2G
Switches: 650 V SiC 60 mOhm TO-247-4 (Kelvin source). Dead time 0.5 us.

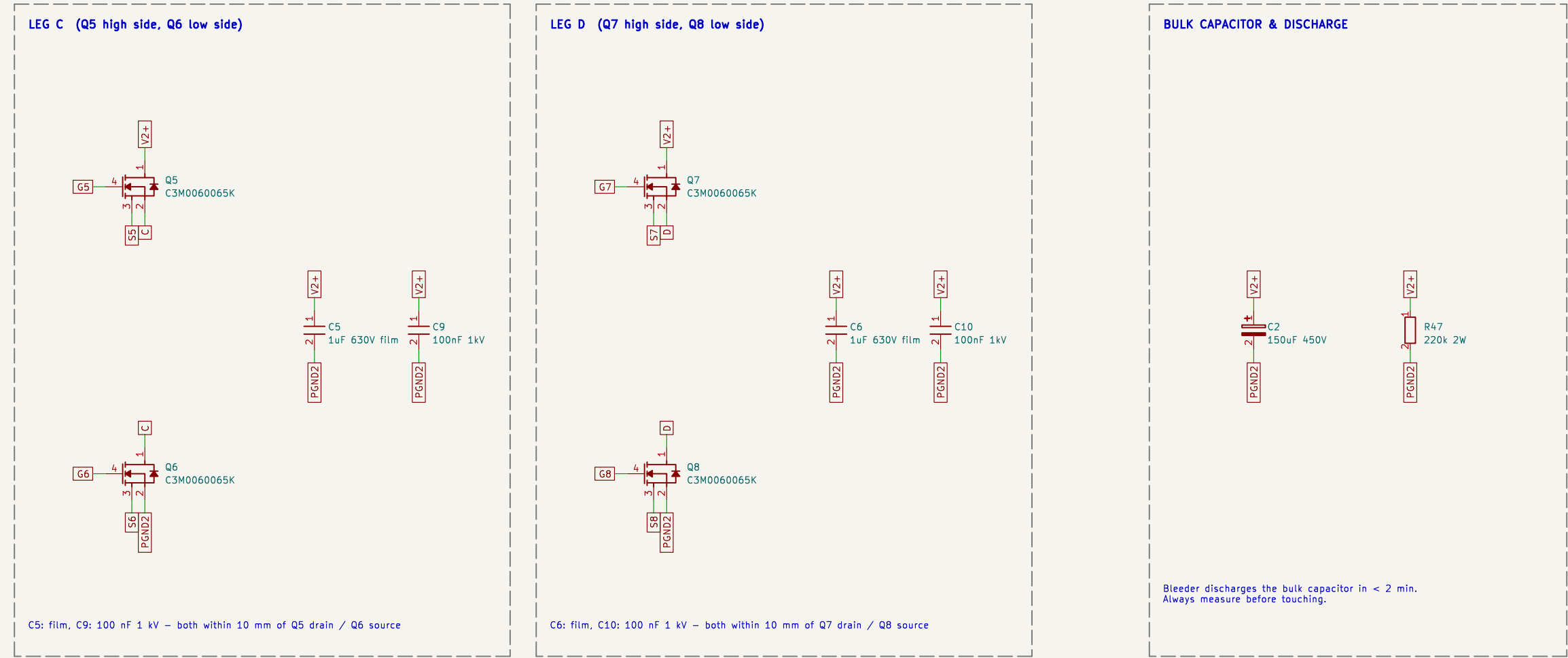
DEVIATIONS FROM THE PLAN (see README): 8 gate supplies instead of 6 – with Kelvin-source MOSFETs each
low-side driver must return to its own Kelvin pin, so the two low sides cannot share one supply.
Current sensors: LEM LTSR 25-NP (single +5 V) instead of LAH 25-NP (needs +/-15 V).

Rev 3.0: hierarchical – Bridge1, Bridge2, GateDrive, Sensing, Connectors		
V1 400 V V2 350 V 3.3 kW fs 20 kHz n 8:7 Lk 291 uH dead time 0.5 us		
Anuj Kumar (23EE039), Ankush (23EE033)		
B.Tech Major Project–I (2026–27): 3.3 kW bidirectional DAB for EV charging with V2G		
Delhi Technological University – Dept. of Electrical Engineering		
Sheet: /		
File: DAB_3k3W.kicad_sch		
Title: DAB_3k3W – 3.3 kW DAB power board (top level)		
Size: A3	Date: 2026-10-03	Rev: 3.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/6

Bridge 1 – primary full bridge (400 V DC bus)



Bridge 2 – secondary full bridge (350 V battery)

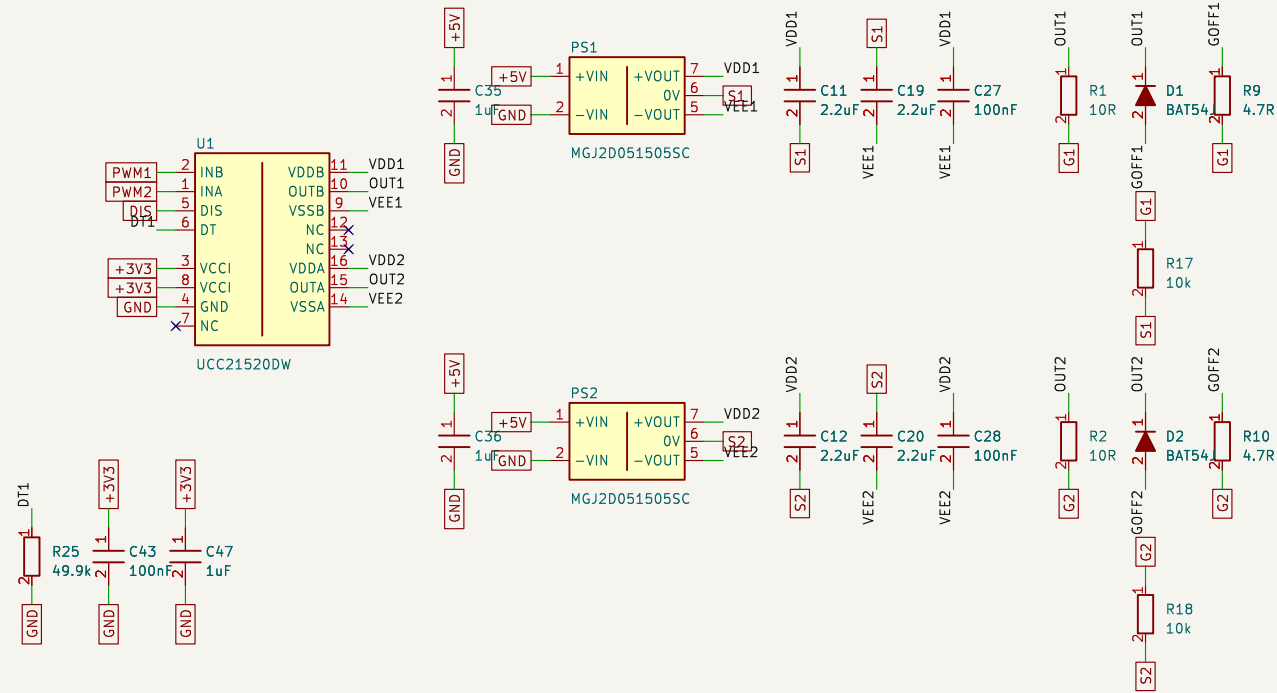


Rev 3.0: lab prototype per KiCad build plan (3 isolated domains)		
V1 400 V V2 350 V 3.3 kW fs 20 kHz n 8:7 Lk 291 uH dead time 0.5 us		
Anuj Kumar (23EE039), Ankush (23EE033)		
B.Tech Major Project–I (2026–27): 3.3 kW bidirectional DAB for EV charging with V2G		
Delhi Technological University – Dept. of Electrical Engineering		
Sheet: /Bridge2/		
File: Bridge2.kicad_sch		
Title: Bridge 2 – secondary full bridge (350 V battery)		
Size: A3	Date: 2026–10–03	Rev: 3.0
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 3/6

Isolated gate drive – 4 x UCC21520, 8 x isolated supplies

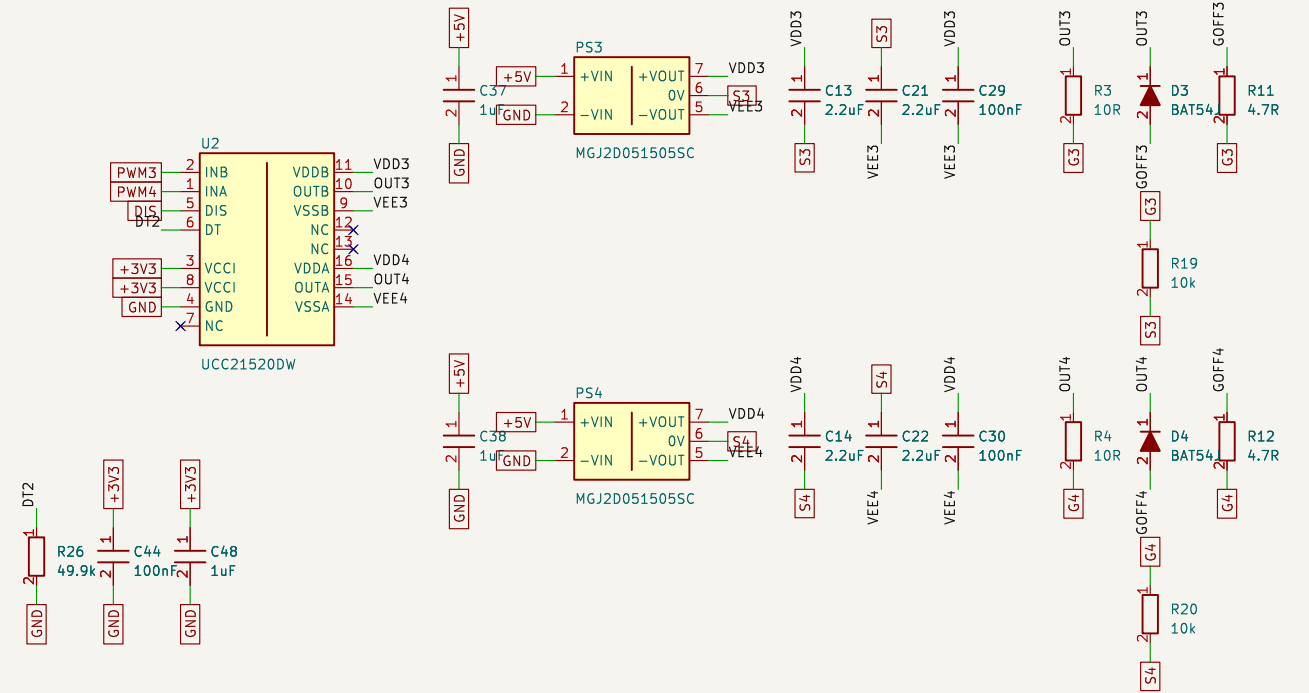
R29: DIS pull-up – all drivers stay disabled until the controller pulls DIS low.

U1: LEG A – ch.B -> Q1 (high side), ch.A -> Q2 (low side)



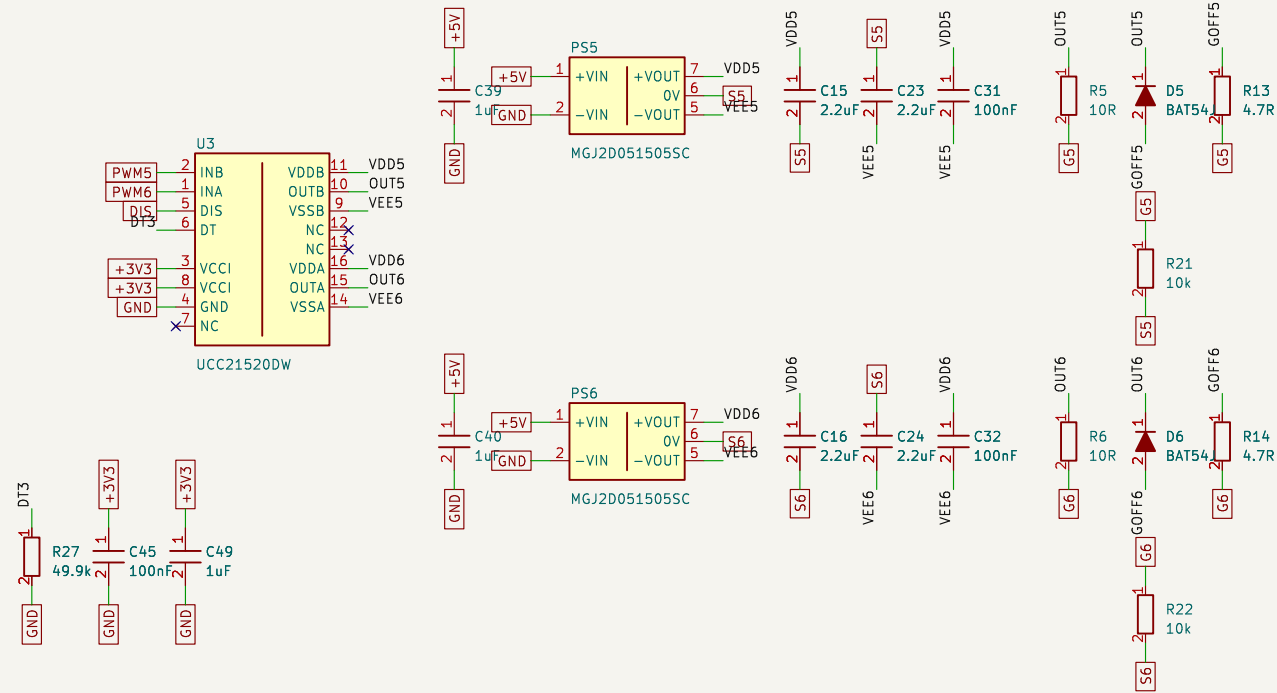
PS1 / PS2: +15 V / -5 V, each referenced to its own Kelvin source (S1, S2)

U2: LEG B – ch.B -> Q3 (high side), ch.A -> Q4 (low side)



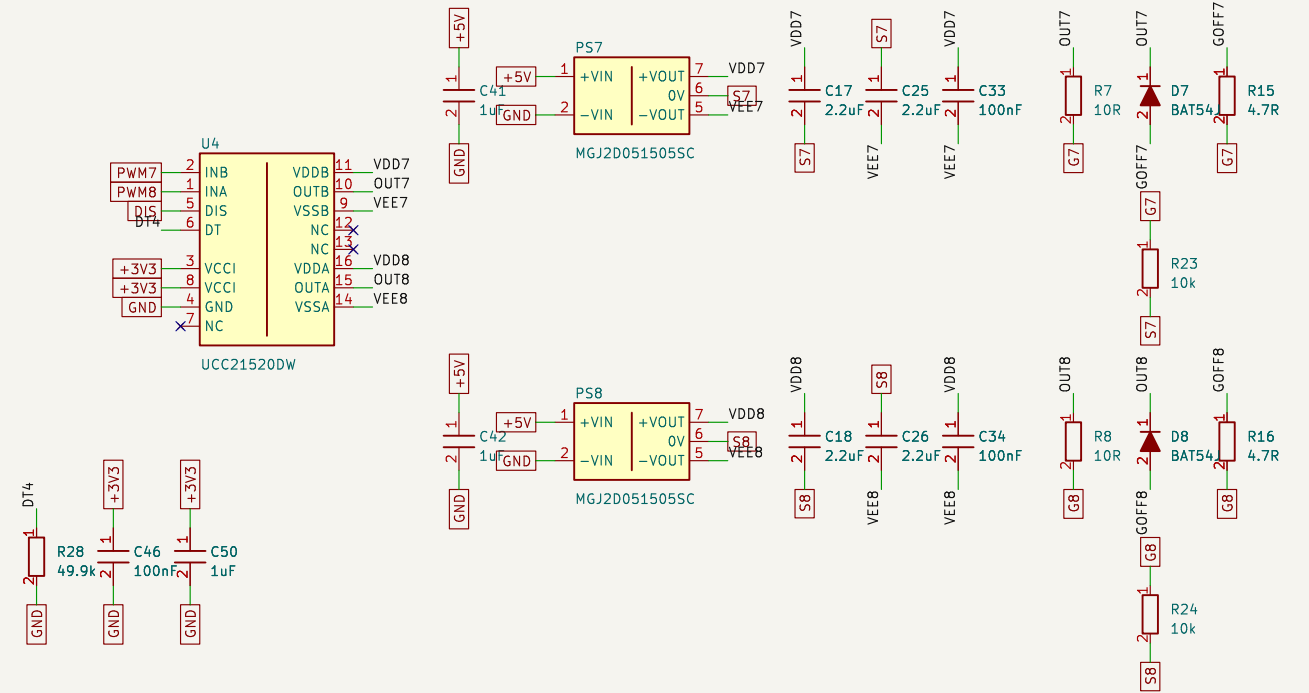
PS3 / PS4: +15 V / -5 V, each referenced to its own Kelvin source (S3, S4)

U3: LEG C – ch.B -> Q5 (high side), ch.A -> Q6 (low side)



PS5 / PS6: +15 V / -5 V, each referenced to its own Kelvin source (S5, S6)

U4: LEG D – ch.B -> Q7 (high side), ch.A -> Q8 (low side)



PS7 / PS8: +15 V / -5 V, each referenced to its own Kelvin source (S7, S8)

1	2	3	4	5	6	7	8
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U7 V15 AMC1



U7 V15 AMC1

LTSR 25-NP: 1-turn primary (25 A). $V_{out} = 2.5\text{ V} + 0.625\text{ V} \times I_{25\text{ A}}$ R divider scales the full 0.375..4.625 V span into the 0..3.3 V ADC range.	LTSR 25-NP: 1-turn primary (25 A). $V_{out} = 2.5\text{ V} + 0.625\text{ V} \times I_{25\text{ A}}$ R divider scales the full 0.375..4.625 V span into the 0..3.3 V ADC range.
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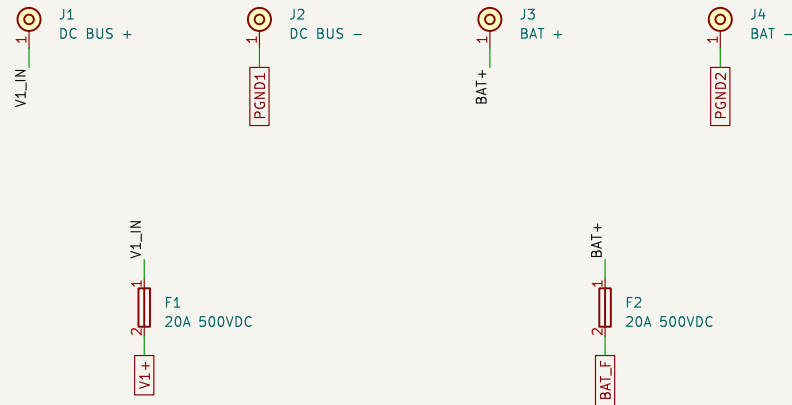
Rev 3.0: Lab prototype per KiCad build plan (3 isolated domains) V1 400 V V2 350 V 3.3 kW fs 20 kHz n 8:7 Lk 291 uH dead time 0.5 us Anuj Kumar (23EE039), Ankush (23EE033) B.Tech Major Project-I (2026-27): 3.3 kW bidirectional DAB for EV charging with V2G Delhi Technological University – Dept. of Electrical Engineering Sheet: /Sensing/ File: Sensing.kicad_sch
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Title: Isolated V/I sensing – AMC1311 + LEM LTSR 25-NP

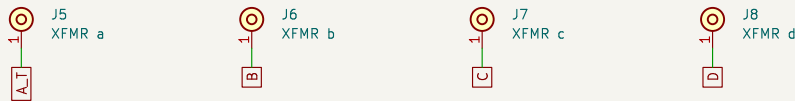
Size: A3	Date: 2026-10-03	Rev: 3.0
KicCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

Power terminals, fuses, control header, logic supply

POWER TERMINALS & FUSES (M4 ring-lug pads)

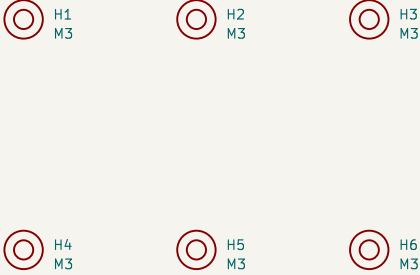


TRANSFORMER / Lk TERMINALS (off-board magnetics)



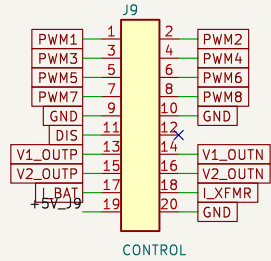
Off-board: HF transformer 8:7 (ETD59 / E65) and Lk = 291 uH (or built into the transformer leakage). Keep a/b and c/d leads twisted.

MOUNTING



H1-H4 board mounting, H5-H6 heatsink mounting (5 mm spacers, heatsink >= 8 mm from control area)

CONTROL HEADER J9 (to C2000 LaunchPad)



PWM1..8 drive Q1..Q8 (diagonal pairs set in firmware: PG1 -> Q1+Q4, PG2 -> Q2+Q3, PG3 -> Q5+Q8, PG4 -> Q6+Q7). DIS: pull low to enable all drivers. Pin 12 reserved (FAULT). Dead time 0.5 us in firmware; UCC21520 DT pins give a 500 ns hardware floor.

LOGIC SUPPLY (+5 V OR-ing, 3.3 V LDO)

